# Exit Ticket · reflect and retrieve

## Exit Ticket

A hash output can normally be decrypted to recover the exact original password.

## Use key vocabulary from memory

1. Explain hash function in your own words.

2. Explain digest in your own words.

3. Explain collision in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 133 · Hashing for Integrity, Passwords and Data Lookup

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

How does hybrid encryption usually protect bulk web traffic?

### Q2

What are the outputs of a half adder when A=1 and B=1?

### Q3

In Big O notation, what is binary search worst-case complexity?

### Q4

When does a linear search stop?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Compute a digest — A cryptographic hash produces a reproducible fingerprint of file content.

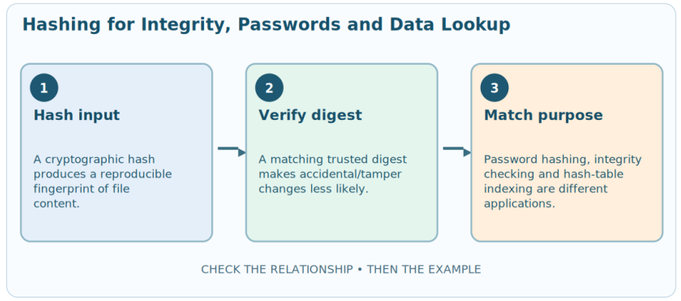
## Core idea 2

Compare for integrity — A matching trusted digest makes accidental/tamper changes less likely.

## Core idea 3

Choose correct use — Password hashing, integrity checking and hash-table indexing are different applications.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Evaluate the claim 'a hash can decrypt a password and proves a file came from its claimed author'. [8]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

State three different uses of hashing with one correct explanation each. [6]

## Question 2

Explain what a hash collision is and why it matters. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

hash function — Maps input data to a fixed-size or bounded-size result.

digest — Output of a cryptographic hash function.

collision — Two distinct inputs produce the same hash result.

salt — Random value combined with a password before secure password hashing.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.

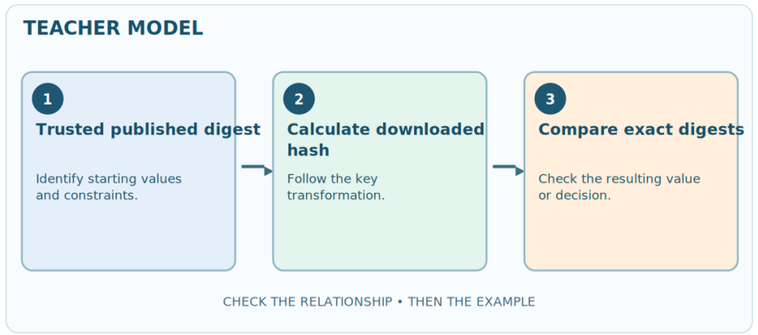
## Follow the worked method

1. Trusted published digest — Find the given data and name the required result.

2. Calculate downloaded hash — Explain why this intermediate stage is necessary.

3. Compare exact digests — Calculate SHA-256 digest of received file and compare with the digest from a trusted source. If different, file changed or was corrupted. If equal, this strongly supports matching content, assuming trusted digest and secure algorithm; it does not by itself prove publisher identity.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Choose random salt
Find the given data and name the required result.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.